

39.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 58



Bug means

- A. A logical error in a program
- B. A difficult syntax error in a program
- C. Documenting programs using an efficient documentation tool
- D. All of the above

nielit2017july-scientistb-cs programming

Answer key

39.0.2 NIELIT 2017 July Scientist B (IT) - Section B: 37



Consider the following C declaration

```
struct
{
    short s[5];
    union
    {
        float y;
        long z;
    }u;
}t;
```

Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes respectively. The memory requirement for variable t ignoring alignment considerations, is

- A. 22 bytes
- B. 14 bytes
- C. 18 bytes
- D. 10 bytes

nielit2017july-scientistb-it programming

Answer key

39.0.3 NIELIT 2017 July Scientist B (IT) - Section B: 26



The following program is to be tested for statement coverage:

```
begin
if(a==b){S1;exit;}
else if (c==d){S2;}
else{S3;exit;}
S4;
end
```

The test cases $T1$, $T2$, $T3$ and $T4$ given below are expressed in terms of the properties satisfied by the values of variables a , b , c and d . The exact values are not given.

$T1$: a , b , c and d are all equal

$T1$: a , b , c and d are all distinct

$T3$: $a = b$ and $c \neq d$

$T4$: $a \neq b$ and $c = d$

Which of the test suites given below ensures coverage of statements $S1$, $S2$, $S3$ and

S4?

A. T_1, T_2, T_3

B. T_2, T_4

C. T_3, T_4

D. T_1, T_2, T_4

nielit2017july-scientistb-it programming

Answer key

39.0.4 NIELIT 2016 MAR Scientist B - Section C: 45



Control structures include

A. iteration

B. rendezvous statements

C. exception statements

D. all of these

nielit2016mar-scientistb programming

Answer key

39.1 Array of Pointers (4)

39.1.1 Array of Pointers: NIELIT 2016 DEC Scientist B (IT) - Section B: 11



What is the meaning of following declaration?

```
int(*p[7])();
```

A. p is pointer to function

B. p is pointer to such function which return type is array

C. p is array of pointer to function

D. p is pointer to array of function

nielit2016dec-scientistb-it programming-in-c pointers array-of-pointers

Answer key

39.1.2 Array of Pointers: NIELIT 2017 DEC Scientific Assistant A - Section B: 42

Which of the following is illegal declaration in C language?



A. `char*str="Raj is a Research scholar";`

B. `char str[25]="Raj is a Research scholar";`

C. `char str[40]="Raj is a Research scholar";`

D. `char[]str="Raj is a Research scholar";`

nielit2017dec-assistanta programming-in-c pointers array-of-pointers

Answer key

39.1.3 Array of Pointers: NIELIT 2017 DEC Scientist B - Section B: 44



If x is a one dimensional array, then

- A. $*(x + i)$ is same as $*(&x[i])$
C. $*(x + i)$ is same as $*x[i]$

- B. $&x[i]$ is same as $x + i - 1$
D. $*(x + i)$ is same as $*x + i$

nielit2017dec-scientistb programming-in-c array-of-pointers

Answer key

39.1.4 Array of Pointers: NIELIT 2017 July Scientist B (IT) - Section B: 35



Assume that float takes 4 bytes, predict the output of following program.

```
#include<stdio.h>
int main()
{
    float arr[5]={12.5,10.0,13.5,90.5,0.5};
    float *ptr1=&arr[0];
    float *ptr2=ptr1+3;
    printf("%f",*ptr2);
    printf("%d",ptr2-ptr1);
    return 0;
}
```

- A. 90.500000 3
C. 10.000000 12
B. 90.500000 12
D. 0.500000 3

nielit2017july-scientistb-it programming-in-c pointers array-of-pointers

Answer key

39.2 Data Structures (1)

39.2.1 Data Structures: NIELIT 2018-41



_____ is used to convert from recursive to iterative implementation of an algorithm

- A. Array
B. Tree
C. Stack
D. Queue

nielit-2018 data-structures

Answer key

39.3 Dynamic Scoping (1)

39.3.1 Dynamic Scoping: NIELIT 2016 DEC Scientist B (IT) - Section B: 60



How will you free the allocated memory?

- A. remove(var-name)
C. delete(var-name)
B. free(var-name)
D. malloc(var-name)

nielit2016dec-scientistb-it programming-in-c dynamic-scoping

Answer key

39.4 Functions (1)

39.4.1 Functions: NIELIT 2016 DEC Scientist B (IT) - Section B: 25



The keyword used to transfer control from a function back to the calling function is:

A. switch

B. go to

C. go back

D. return

nielit2016dec-scientistb-it programming functions

Answer key

39.5

Output (13)

39.5.1 Output: NIELIT 2016 DEC Scientist B (CS) - Section B: 13



What will be output if you will compile and execute the following C code?

```
void main()
{
    printf("%d",sizeof(5.2));
}
```

A. 4

B. 8

C. 2

D. 16

nielit2016dec-scientistb-cs programming-in-c output

Answer key

39.5.2 Output: NIELIT 2016 DEC Scientist B (CS) - Section B: 8



What will be output if you will compile and execute the following C code?

```
void main(){
char c=125;
c=c+10;
printf("%d",c);
}
```

A. 135

B. 115

C. -121

D. -8

nielit2016dec-scientistb-cs programming-in-c output

Answer key

39.5.3 Output: NIELIT 2016 MAR Scientist B - Section C: 15



Output of the following loop is

```
for(putchar('c');putchar
('a');putchar('r'))
    putchar('t');
```

A. a syntax error.

B. cartrt.

C. catrat.

D. catratratratrat...

nielit2016mar-scientistb programming-in-c output

Answer key

39.5.4 Output: NIELIT 2017 July Scientist B (IT) - Section B: 32



Output of following program

```
#include<stdio.h>
int main()
```

```

{
int i=5;
printf("%d %d %d", i++,i++,i++);
return 0;
}

```

- A. 7 6 5 B. 5 6 7 C. 7 7 7 D. Compiler Dependent

nielit2017july-scientistb-it programming-in-c output

Answer key 

39.5.5 Output: NIELIT 2017 July Scientist B (IT) - Section B: 33



Output of following program?

```

#include<stdio.h>
void dynamic(int s,...)
{
printf("%d",s);
}
int main()
{
dynamic(2,4,6,8);
dynamic(3,6,9);
return 0;
}

```

- A. 2 3 B. Compiler Error C. 4 3 D. 3 2

nielit2017july-scientistb-it programming-in-c output

Answer key 

39.5.6 Output: NIELIT 2017 July Scientist B (IT) - Section B: 36



Assume that size of an integer is 32 bit. What is the output of following ANSI C program?

```

#include<stdio.h>
struct st
{
int x;
static int y;
};
int main()
{
printf("%d",sizeof(struct st));
return 0;
}

```

- A. 4 B. 8 C. Compiler Error D. Runtime Error

nielit2017july-scientistb-it programming-in-c output

Answer key 

39.5.7 Output: NIELIT 2017 July Scientist B (IT) - Section B: 38



Output of the following program?

```
#include<stdio.h>
struct st
{
    int x;
    struct st next;
};
int main()
{
    struct st temp;
    temp.x=10;
    temp.next=temp;
    printf("%d",temp.next,x);
    return 0;
}
```

- A. Compiler Error B. 10 C. Runtime Error D. Garbage Value

nielit2017july-scientistb-it programming-in-c output

Answer key 

39.5.8 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 6



The question is based on the following program fragment.

```
f(int Y[10],int x){
    int u,j,k;
    i=0;j=9;
    do{
        k=(i+j)/2;
        if(Y[k] < x) i=k; else j=k;
    } while(Y[k]!=x) && (i<j));
    if (Y[k]==x) printf("x is in the array.");
    else printf("x is not in the array.");
}
```

On which of the following contents of 'Y' and 'x' does the program fail?

- A. Y is [1 2 3 4 5 6 7 8 9 10] and $x < 10$
 B. Y is [1 3 5 7 9 11 13 15 17 19] and $x < 1$
 C. Y is [2 2 2 2 2 2 2 2 2 2] and $x > 2$
 D. Y is [2 4 6 8 10 12 14 16 18 20] and $2 < x < 20$ and 'x' is even

nielit2017oct-assistanta-cs programming-in-c output

Answer key 

39.5.9 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 11



What will be the output of following?

```
main()
{
    Static int a = 3;
    Printf("%d",a--);
    If(a)
        main();
}
```

- A. 3
- B. 3 2 1
- C. 3 3 3
- D. Program will fall in continuous loop and print 3

nielit2017oct-assistanta-cs programming-in-c output

Answer key 

39.5.10 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 3



The following program fragment prints

```
int i = 5;
do { putchar(i+100); printf("%d", i--); }
while(i);
```

- A. i5h4g3f2el
- B. 14h3g2f1e0
- C. An error message
- D. None of the above

nielit2017oct-assistanta-cs programming-in-c output

Answer key 

39.5.11 Output: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 1



What will be the output of following?

```
main()
{
    Static int a = 3;
    Printf("%d",a--);
    If(a)
        main();
}
```

- A. 3
- B. 3 2 1
- C. 3 3 3
- D. Program will fall in continuous loop and print 3

Answer key

39.5.12 Output: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 1



The question is based on the following program fragment.

```
f(int Y[10], int x){
    int u, j, k;
    i=0; j=9;
    do{
        k=(i+j)/2;
        if(Y[k] < x) i=k; else j=k;
    } while((Y[k]!=x) && (i<j));
    if (Y[k]==x) printf("x is in the array.");
    else printf("x is not in the array.");
}
```

On which of the following contents of 'Y' and 'x' does the program fail?

- A. Y is [1 2 3 4 5 6 7 8 9 10] and $x < 10$
- B. Y is [1 3 5 7 9 11 13 15 17 19] and $x < 1$
- C. Y is [2 2 2 2 2 2 2 2 2 2] and $x > 2$
- D. Y is [2 4 6 8 10 12 14 16 18 20] and $2 < x < 20$ and 'x' is even

Answer key

39.5.13 Output: NIELIT 2021 Dec Scientist B - Section B: 21



What will be the output of the following 'C' program?

```
void count(int n)
{
    static int d = 1;
    printf(" %d", n);
    printf(" %d", d);
    d++;
    if (n>1) count (n - 1);
    printf(" %d", d);
}
void main()
{
    count(3);
}
```

A. 3 1 2 2 1 3 4 4 4

B. 3 1 2 1 1 1 2 2 2

C. 3 1 2 2 1 3 4

D. 3 1 2 1 1 1 2

nielit-2021-it-dec-scientistb programming-in-c output

Answer key

39.6

Pointers (3)

39.6.1 Pointers: NIELIT 2016 MAR Scientist B - Section C: 14



Prior to using a pointer variable it should be

- A. declared.
- B. initialized.
- C. both declared and initialized.
- D. none of these.

nielit2016mar-scientistb programming-in-c pointers

Answer key

39.6.2 Pointers: NIELIT 2017 July Scientist B (IT) - Section B: 34



Output of following program?

```
#include<stdio.h>
int main()
{
    int *ptr;
    int x;
    ptr=&x;
    *ptr=0;
    printf("x=%d\n",x);
    printf("*ptr=%d\n",*ptr);
    *ptr+=5;
    printf("x=%d\n",x);
    printf("*ptr=%d\n",*ptr);
    (*ptr)++;
    printf("x=%d\n",x);
    printf("*ptr=%d\n",*ptr);
    return 0;
}
```

- A. $x = 0$
 $*ptr = 0$
 $x = 5$
 $*ptr = 5$
 $x = 6$
 $*ptr = 6$
- B. $x =$ garbage value
 $*ptr = 0$
 $x =$ garbage value
 $*ptr = 5$
 $x =$ garbage value
 $*ptr = 6$
- C. $x = 0$
 $*ptr = 0$
 $x = 5$
 $*ptr = 5$

- x =garbage value
 $*ptr$ =garbage value
 D. $x = 0$
 $*ptr = 0$
 $x = 0$
 $*ptr = 0$
 $x = 0$
 $*ptr = 0$

nielit2017july-scientistb-it programming-in-c pointers

Answer key

39.6.3 Pointers: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 29



Consider the following declaration.

```
int a, *b = &a, **c = &b;
a = 4;
**c = 5;
```

If the statement

```
b = (int *)**c
```

Is appended to the above program fragment then

- A. Value of b becomes 5
 B. Value of b will be the address of c
 C. Value of b is unaffected
 D. None of these

nielit2017oct-assistanta-it programming pointers

Answer key

39.7 Programming In C (10)

39.7.1 Programming In C: NIELIT 2016 DEC Scientist B (IT) - Section B: 10



The maximum combined length of the command-line arguments including the spaces between adjacent arguments is:

- A. 128 characters
 B. 256 characters
 C. 67 characters
 D. It may vary from one Operating System to another

nielit2016dec-scientistb-it programming-in-c

Answer key

39.7.2 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 11



What is the correct way to round off x , a float to an int value?

- A. $y = (\text{int})(x + 0.5)$
 B. $y = \text{int}(x + 0.5)$
 C. $y = (\text{int})x + 0.5$
 D. $y = (\text{int})(\text{int})x + 0.5$

Answer key **39.7.3 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 13**

What error would the following function give on compilation?

```
f(int a, int b)
{
    int a;
    a=20;
    return a;
}
```

- A. Missing parenthesis is *return* statement.
 B. Function should be defined as `int f(int a, int b)`
 C. Redclaration of *a*.
 D. None of these.

Answer key **39.7.4 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 16**

If space occupied by two strings s_1 and s_2 in 'C' are respectively m and n , then space occupied by string obtained by concatenating s_1 and s_2 is always

- A. less than $m + n$
 B. equal to $m + n$
 C. greater than $m + n$
 D. none of these

Answer key **39.7.5 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 29**

An external variable

- A. is globally accessible by all functions
 B. has a declaration "extern" associated with it when declared within a function
 C. will be initialized to 0 if not initialized
 D. all of these

Answer key **39.7.6 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 57**

If initialization is a part of declaration of a structure, then storage class can be

- A. automatic B. register C. static D. anything

Answer key 

39.7.7 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 58



For x and y are variables as declared below $double\ x = 0.005, y = -0.01$; What is the value of $\text{ceil}(x + y)$, where ceil is a function to compute ceiling of a number?

- A. 1 B. 0 C. 0.005 D. 0.5

nielit2016mar-scientistc programming-in-c

Answer key

39.7.8 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 59



In C programming language, if the first and the second operands of operator $+$ are of types `int` and `float`, respectively, the result will be of type

- A. `int` B. `float` C. `char` D. `long int`

nielit2016mar-scientistc programming-in-c

Answer key

39.7.9 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 60



What will be the value of x and y after execution of the following statement(C language)

```
n = 5; x = n++; y = -x;
```

- A. 5, -4 B. 6, -5 C. 6, -6 D. 5, -5

nielit2016mar-scientistc programming-in-c

Answer key

39.7.10 Programming In C: NIELIT 2017 July Scientist B (CS) - Section B: 5



Following is C like Pseudo code of a function that takes a number as an argument, and uses a stack S to do processing.

```
void fun(int n)
{
    Stack S; // Say it creates an empty stack S
    while(n > 0)
    {
        // This line pushes the value of n%2 to stack S;
        Push(&S, n%2);
        n = n/2;
    }
    // Run while Stack S is not empty
    while(!is Empty(&S))
        printf("%d", pop(&S)); // pop an element from S and print it
}
```

What does the above function do in general ?

- A. Prints binary representation of n in reverse order.

- B. Prints binary representation of n .
- C. Prints the value of $\log n$.
- D. Prints the value of $\log n$ in reverse order.

nielit2017july-scientistb-cs programming-in-c

Answer key 

39.8 Recursion (2)

39.8.1 Recursion: NIELIT 2021 Dec Scientist B - Section B: 32



Consider the following C function :

```
int f(int n) {
    static int i = 1;
    if (n >= 5) return n;
    n=n + i; i++;
    return f(n); }
```

The value returned by $f(1)$ is :

- A. 5
- B. 6
- C. 7
- D. 8

nielit-2021-it-dec-scientistb programming-in-c recursion functions

Answer key 

39.8.2 Recursion: NIELIT 2021 Dec Scientist B - Section B: 72



What is the value of $f(4)$ using the following C code:

```
int f(int k){
    if(k<3)
        return k;
    else
        return f(k-1) * f(k-2) + f(k-3);
}
```

- A. 5
- B. 6
- C. 7
- D. 8

nielit2021dec-scientistb programming-in-c recursion recurrence-relation data-structures

Answer Keys

39.0.1	A	39.0.2	C	39.0.3	D	39.0.4	D	39.1.1	C
39.1.2	D	39.1.3	A	39.1.4	A	39.2.1	C	39.3.1	B
39.4.1	D	39.5.1	B	39.5.2	C	39.5.3	D	39.5.4	D
39.5.5	A	39.5.6	C	39.5.7	A	39.5.8	C	39.5.9	B
39.5.10	A	39.5.11	B	39.5.12	C	39.5.13	A	39.6.1	C

39.6.2	A
39.7.4	A
39.7.9	D

39.6.3	A
39.7.5	D
39.7.10	B

39.7.1	D
39.7.6	C
39.8.1	C

39.7.2	A
39.7.7	B
39.8.2	A

39.7.3	C
39.7.8	B